Do Upper-Ocean Conditions Improve Prediction of Marine Heatwave Severity in the Northwest Atlantic?

GEOL599 Final Project Proposal

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1 Background

Marine heatwaves (MHWs) are discrete periods of anomalously warm sea surface temperature (SST) that can affect marine ecosystems, fisheries, aquaculture, and coastal communities [*Hobday et al.*, 2016; *Oliver et al.*, 2018]. A widely used MHW definition identifies events when daily SST exceeds a seasonally varying threshold for at least five consecutive days [*Hobday et al.*, 2016]. Global observations show that MHWs have become more frequent, intense, and persistent over the last century, consistent with a warming ocean baseline [*Oliver et al.*, 2018]. However, the severity of an individual MHW cannot be understood from SST alone; its evolution may also reflect the local circulation, water-mass properties, and vertical structure of the upper ocean. The Northwest (NW) Atlantic is a particularly suitable region in which to investigate these relationships. Within the proposed study domain, the Gulf Stream (GS) creates strong and highly variable SST, sea surface salinity (SSS), and sea surface height (SSH) gradients. Shifts in GS position and associated SSH anomalies have been linked to downstream changes in the temperature and salinity of NW Atlantic shelf waters [*Gonçalves Neto et al.*, 2021]. In addition, GS meanders can shed warm and cold core rings (WCRs, CCRs)1 which redistribute heat across the region [*Silver et al.*, 2021]. Thus, local SSH and SSS anomalies may provide indicators of GS proximity, meander state, or eddy-related water mass influence that are not represented by local SST.

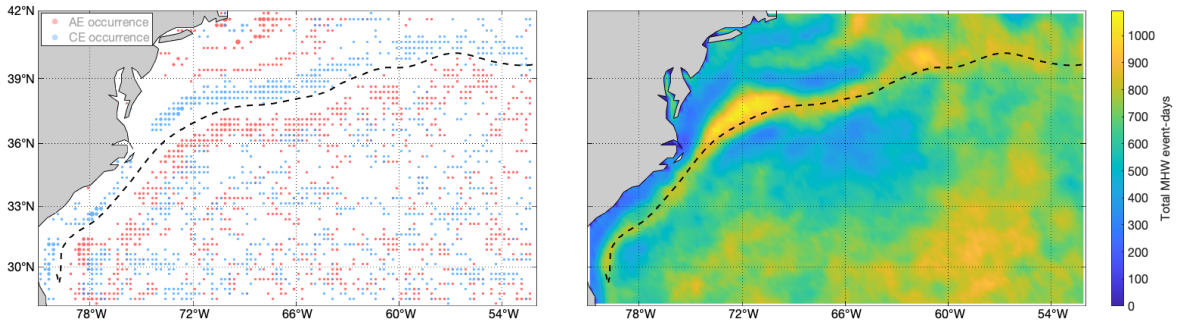


Figure 1: (Left) Spatial distribution of eddy occurrence; grid cells in the upper 20% of AE occurrence are shown in red and grid cells in the upper 20% of CE occurrence are shown in blue. (Right) MHW event-days during 2016–2024. The dashed black line indicates the climatological GS position.

Mixed layer depth (MLD) is also physically relevant because it influences the volume of water over which surface heating and cooling are distributed. For example, the notable summer 2023 North Atlantic MHW was associated with anomalously shallow mixed layers and weak winds over much of the basin [*England et al.*, 2025]. These results motivate including MLD in the feature set along with SST, SSS, and SSH anomalies.

2 Objective

This project evaluates the predictive value of antecedent upper-ocean conditions rather than attempting to attribute individual MHWs to a single mechanism. By comparing models that use only season, location, and SST against otherwise identical models that also include SSS, SSH, and MLD, the analysis will directly test whether these additional satellite- and reanalysis-based variables provide useful information for predicting MHW severity in the Northwest Atlantic.

3 Data and Preprocessing Methods

The dataset will comprise MHW events detected during 2016–2024 at OISST grid cells within 28–42°N and 81–51°W. Events and their metrics will be identified using the MATLAB MHW Toolbox following the definition of *Hobday et al.* [2016] [*Zhao and Marin*, 2019a, b]. Each observational unit will be one selected MHW onset at one grid cell. The final sample size will be reported after collocating all predictor datasets and selecting appropriate events. Two continuous targets will be modeled separately: (1) peak MHW intensity in °C above the seasonally varying climatological threshold, and (2) event duration in days. Predictors will be restricted to information available before the onset date. Daily 0.25° SST observations are obtained from the National Oceanic and Atmospheric Administration (NOAA) Optimum Interpolation Sea Surface Temperature product (OISST, version 2.1; 1982–2024) [*Huang et al.*, 2021; *NOAA National Centers for Environmental Information*, 2023]. The SSS product used is from National Aeronautics and Space Administration (NASA) Soil Moisture Active Passive (SMAP, version 6.0; 2016–2024) obtained from the Remote Sensing Systems (RSS) which has 70 km spatial resolution and an 8-day running mean from L3 gridded data [*Meissner et al.*, 2024]. SSH fields are derived from the Copernicus Marine Environment Monitoring Service (CMEMS) multi-mission altimeter product [*Pujol et al.*, 2016; *Copernicus Marine Environment Monitoring Service*, 2023a]. These Level-4 mapped fields are generated through optimal interpolation of along-track altimeter measurements and provided at 0.25° spatial resolution. MLD is obtained from the Global Ocean Reanalysis and Simulation GLORYS12V1 [*Lellouche et al.*, 2018; *Copernicus Marine Environment Monitoring Service*, 2023b], which assimilates satellite and *in situ* observations into the NEMO ocean model and provides daily-averaged output at 1/12° horizontal resolution with 50 vertical levels, which was downscaled to 0.25° horizontal resolution to match the satellite observations. For each variable, I will calculate antecedent summaries over 2-, 7-, 30-, and 60-day windows, including mean conditions, variability, and recent trend.

4 Proposed Modeling Methods

4.1 Exploratory Analysis

I will first examine the spatial and temporal distribution of detected MHWs. A central concern is that many nearby grid-cell events may belong to one larger contiguous warming event and therefore should not be treated as independent observations. I will identify candidate events that are separated by at least 30 days from preceding and subsequent events at the same grid cell. I will then retain at most one representative event per month within each of three physically distinct regions: a coastal/northwest of GS region, a GS path region, and a southeastern/offshore region. Using this method, the selected MHW datatset will contain 36 events per year (3 per month). This sampling strategy reduces repeated sampling of the same large scale event while retaining examples influenced by differing regional processes.

4.2 Feature Sets and Models

The central experiment will compare two feature sets within two model frameworks:

1. **SST-only feature set:** latitude, longitude, and antecedent SST anomaly metrics. This set represents prediction using the seasonal cycle (represented by the SST anomaly) and location.
2. **Full upper-ocean feature set:** latitude, longitude, antecedent SST anomaly, SSS anomaly, SSH anomaly, and MLD metrics. This set tests whether additional upper-ocean variables provide predictive information beyond SST.

For each target and feature set, two model structures will be fitted: (1) a linear regression (LR) will provide an interpretable baseline, and (2) an XGBoost regression model will test whether more flexible nonlinearities and interactions improve prediction. Random forest regression may be evaluated as an exploratory comparison depending on the results of XGBoost, but XGBoost will be the primary tree-based model because its boosting framework may better support future year prediction.

4.3 Training, Validation, and Testing

To evaluate predictive capability, I will reserve 2023–2024 as a final test set. Events from 2016–2021 will be used for model fitting, and 2022 will be used for model selection and hyperparameter tuning. As a sensitivity analysis, I will also use leave-one-year-out validation across the pre-test years. This design is more realistic than a random train/test split because it prevents the model from using information from temporally adjacent events to predict later events. The same selected event sample and year-based partitions will be applied to all models and both feature sets.

4.4 Evaluation Metrics and Interpretation

Models will be evaluated on the withheld 2023–2024 data using mean absolute error (MAE), root mean square error (RMSE), and R^2. The primary evidence for the scientific question will be whether the full upper ocean feature set reduces test set MAE and RMSE relative to the SST-only feature set within the same model family. Comparisons across LR and XGBoost will show whether any benefit of the additional variables depends on linear, smooth nonlinear, or more complex interacting relationships. I will interpret model behavior using LR coefficients and XGBoost permutation importance and accumulated local effect plots. Error analysis will assess whether performance differs by region, season, or target severity.

5 Broader Impacts, Limitations, and Fallback Plan

Better understanding of the preconditions associated with intense or persistent MHWs could support the interpretation of satellite observations and eventually contribute to regional risk monitoring. Key limitations are the short 2016–2024 overlap period, spatial dependence among MHW events, and different native resolutions among products. The selected event strategy and withheld year test are meant to reduce (but cannot eliminate) dependence and sampling limitations. If collocation or missingness substantially reduces the sample size, the fallback analysis will retain the SST-only and reduced full feature models using only the most complete predictors and 30-day antecedent summaries. Regardless of predictive skill, conclusions will be limited to the sampled NW Atlantic domain and will not be interpreted as causal evidence.

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